

BINDU CHANDRA SHEKAR REDDY

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Professional Summary

Full-stack web developer with one year of experience in HTML, CSS, JavaScript, PHP and Python. Skilled in building responsive web applications, real-time systems and RESTful APIs. Experienced with SQL databases, version control and debugging. Quick to learn, adaptable and a proactive team player focused on delivering efficient, user-friendly solutions and contributing to project success.

Skills

- **Programming:** PHP, Python, Java, C, SQL, JavaScript, Node.js
- **Tools & Platforms:** Docker, Kubernetes, Git, VS Code, Xcode, Google Kubernetes Engine (GKE), Google Colab

Experience

Teleindia Networks Pvt. Ltd

Jun 2023 - Jul 2024

Web Developer

Bengaluru, India

- Developed and maintained responsive web applications using React, PHP and Node.js, applying CSS modules and Jest testing, which improved usability and cross-device performance as reflected in positive user feedback
- Implemented RESTful APIs and optimized MySQL queries, reducing average response time by 30%.
- Collaborated with designers and product teams, using React and PHP to deliver full-stack solutions that met business requirements and enabled on-time release of key features
- Integrated Java-based backend modules with Spring Boot, ensuring seamless data flow between client and server layers, which eliminated data synchronization errors
- Participated in incident management, debugging and resolving production issues to enhance reliability by 20%.

California State University Sacramento

Feb 2024 - Dec 2024

Teaching Assistant

- Assisted students in learning Java fundamentals, focusing on object-oriented design, debugging and coding best practices, which helped them write clearer and more reliable code.
- Conducted lab sessions and gave targeted feedback on assignments through collaborative tools such as Google Colab and Git, which strengthened students' programming skills and deepened their conceptual understanding

Aug 2025 - Present

Instructional Student Assistant

- Guided students in applying object-oriented programming and computer-graphics concepts using Java, helping them complete weekly coding assignments and deepening their ability to create interactive visualizations
- Supported project reviews and lab exercises to enhance understanding of animation, event handling and software design.

Nanorobotics Embed Technologies

Sep 2021 - Oct 2021

Machine Learning Intern

Bengaluru, India

- Developed and tested ANN models in Python with NumPy, Pandas and Matplotlib, achieving reliable predictions that supported downstream embedded analytics
- Programmed Python-based software modules for embedded devices and built ML inference pipelines, enabling real-time analytics for improved device performance.
- Presented findings on ML integration for embedded robotics to the engineering team, informing design decisions and advancing the project roadmap

Projects

NeuroFusionGPT – Multimodal Stress Detection System

- Built a multimodal deep learning model that predicts 5 stress levels by fusing EEG (brain signals) and ECG (heart signals) using a feature-level neural fusion architecture.
- Implemented Transformer-based EEG modeling (16 features) and MLP-based ECG analysis (188 features) to learn physiological stress patterns from 95K+ signal samples.
- Developed an end-to-end ML pipeline including preprocessing, feature scaling, multimodal fusion, inference, and evaluation using accuracy, precision, recall, F1-score and confusion matrix.
- Integrated an AI wellness feedback module that generates personalized stress-management guidance based on predicted stress levels; trained models in Google Colab (GPU) and deployed locally in Python.

ML-based Medical Diagnostics for Infectious Diseases

- Selected for presentation at IEEE SysCon (Systems Conference) for scalability and real-world diagnostic relevance.
- A supervised ML system to predict infection status from epidemiological surveillance data in high-noise, resource-limited settings.
- Engineered novel clinical features such as comorbidity counts and symptom-to-admission intervals to strengthen predictive signals from demographic data.
- Optimized an XGBoost classifier to achieve 0.80 ROC-AUC and 0.73 recall, prioritizing reduced false negatives for public health safety.

IoT-Based Health Monitoring System with Edge Computing

- Developed an iOS frontend for real-time health tracking via connected IoT devices.
- Integrated edge computing for local data processing to reduce latency and improve performance.
- Containerized services using Docker and deployed on Google Kubernetes Engine (GKE) for scalable backend support.

Plant Leaf Disease Detection Using CNN

- Engineered a CNN model using transfer learning (VGG16, ResNet50), achieving ~95% classification accuracy on plant disease datasets.
- Processed and visualized data using Python libraries (NumPy, Pandas, Matplotlib).
- Delivered a real-time classification system enabling early disease detection and preventive action for farmers.

Education

California State University, Sacramento

Aug 2024 - May 2026

Master of Science, Computer Science

Sacramento, CA

KNS Institute of Technology (VTU)

Aug 2018 - Jun 2022

Bachelor of Engineering, Computer Science and Engineering

Bengaluru, India